

5.1. Practica: Diseño de controlador de un sistema de segundo orden

Ecuaciones principales

$$v_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

- Transformada de Laplace

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

- Procedimiento algebraico

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L s + R} \left(L s + \frac{1}{C s} + 3R \right) I_2(s) = \frac{C L s^2 + 3 C R s + 1}{C L s^2 + C R s} I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{C s} \right) I_2(s) = \frac{C R s + 1}{C s} I_2(s)$$

$$V_e(s) = \left[(L s + 2R) \frac{C L s^2 + 3 C R s + 1}{C L s^2 + C R s} - (L s + R) \right] I_2(s)$$

$$= \frac{3 C L R s^2 + (C s C R^2 + L) s + 2 R}{C L s^2 + C R s} I_2(s)$$

- Funcion de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{C R s + 1}{C s}}{\frac{3 C L R s^2 + (C s C R^2 + L) s + 2 R}{C L s^2 + C R s}} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{C L R s^2 + (C R^2 + L) s + R}{3 C L R s^2 + C s C R^2 + L s + 2 R}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{CLR(s)^2 + (CR^2 + L)(s) + R}{3CLR(s)^2 + (5CR^2 + L)(s) + 2R} \right] = \frac{B}{2R}$$

$$e(s) = \lim_{s \rightarrow 0} 1 - 0.5 = 0.5$$

$$e(s) = 0.5 \text{ V}$$

$$0.070312$$

$$0.216936$$

Estabilidad en lazo abierto

$$3CLR s^2 + (5CR^2 + L)s + 2R = 0$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$2274.868$$

$$11374.068$$

$$2200$$

$$4400$$

$$a = 3CLR$$

$$b = 5CR^2 + L$$

$$c = 2R$$

$$R = 2.2 \text{ K}$$

$$C = 470 \text{ nF}$$

$$L = 68 \text{ mH}$$

$$\lambda_{1,2} = \frac{-(5CR^2 + L) \pm \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)}$$

$$\lambda_1 = -0.3868476579$$

$$\lambda_2 = -53921.56415$$

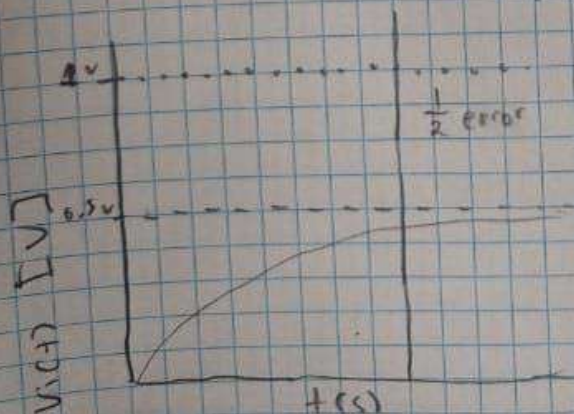
El sistema tiene dos polos negativos y tiene una respuesta sobreamortiguada

Bosquejo

... Entrada del escalon ($v_e = 1$)

--- Estado estacionario (0.5)

— $v(s)$



estado transitorio

estado estacionario

Modelo de ecuaciones Integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir, las corrientes

$$V_e(t) = 2R_1 i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - R_{12} i_2(t)$$

$$i_1(t) = \left[V_e(t) + R_{12} i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R_1}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R_{12} i_2(t) = R_{12} i_2(t) + R_{12} i_2(t) + R_{12} i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + R_{12} i_2(t) \cdot \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R_2}$$

$$i_1(t) = \left[V_e(t) + R_{12} i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R_1}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + R_{12} i_2(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R_2}$$

$$V_s(t) = R_{12} i_2(t) + \frac{1}{C} \int i_2(t) dt$$